Final ORBIT Comparison to Sign-Flipped Einstein-Hilbert Reference

ORBIT

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1 Setup and Conventions

This report compares a Matchete-derived Lagrangian to a reference action

$$\mathcal{L}_{\mathrm{ref}} = \Lambda_{\mathrm{ref}}\sqrt{-g} + s_{\mathrm{EH}}\,\frac{2}{\kappa_{\mathrm{ref}}^2}\sqrt{-g}R \tag{1}$$

with reference coefficients $\Lambda_{\mathrm{ref}} = LambdaRefFinal$, $\kappa_{\mathrm{ref}} = KappaRefFinal$, and $s_{\mathrm{EH}} = -1$. The Matchete input is normalized by setting $hbar = 1$ and discarding all terms containing negative powers of ϵ. If canonical normalization is enabled, the input quadratic sector is rescaled to the target sign $s_{\mathrm{kin}} = -1$ using an extracted symbolic factor Z_h. The raw factor used in the computation is:

```
(M^2*\[Kappa]^2*(1 + Log[\[Mu]bar2/M^2]))/24
```

$$\beta^2 = \frac{s_{\mathrm{kin}}}{Z_h}. \tag{2}$$

2 Reference Support Through Mass Dimension 6

The supported EH plus cosmological sectors are $\{n_h, 0\}$ together with the derivative sectors allowed by the requested mass-dimension cutoff.

3 Reduced Matchete Input

Sector	Input terms	IBP terms	Reduced terms	Coordinates
1, 0	2	2	2	1
2, 0	4	4	4	2
2, 2	10	4	4	4
2, 4	14	5	5	5
3, 0	6	6	6	3
3, 2	42	13	6	10
4, 0	10	10	10	5
4, 2	99	30	25	31

5, 0	14	14	14	7
6, 0	22	22	22	11

4 Reduced Fixed Reference

Sector	Input terms	IBP terms	Reduced terms	Coordinates
1, 0	1	1	1	1
1, 2	2	0	0	0
2, 0	2	2	2	2
2, 2	10	4	4	4
3, 0	3	3	3	3
3, 2	28	13	6	10
4, 0	5	5	5	5
4, 2	66	38	25	31
5, 0	6	6	6	7
6, 0	9	9	9	11

5 Raw Sector-by-Sector Comparison

Sector	Input coords	Reference coords	Raw diff. zero
1, 0	1	1	no
2, 0	2	2	no
2, 2	4	4	yes
3, 0	3	3	no
3, 2	10	10	no
4, 0	5	5	no
4, 2	31	31	no
5, 0	7	7	no
6, 0	11	11	no

The table above reports the unreduced symbolic comparison before the explicit verification substitutions are imposed. The exact-match conclusion below uses the verified branch together with the quadratic completion data.

5.1 Unsupported sectors

None.

5.2 Verification substitutions

```
{Scalar[M] -> M, Scalar[\[Kappa]] -> \[Kappa], Scalar[\[Mu]bar2] -> \[Mu]bar2, Sqrt[1/(M^2*\[Ka
```

5.3 Verified supported-sector match

True

5.4 Quadratic completion

```
<|"R2" -> 0, "Ricci2" -> 0|>
```

```
<|1 -> -3/(10*Scalar[M]^2), 2 -> -3/(20*Scalar[M]^2), 3 -> 3/(10*Scalar[M]^2)|>
```

5.5 Supported-sector matching equations

```
{-4*KappaRefFinal*LambdaRefFinal + 3*Sqrt[6]*M^4*\[Kappa]*Sqrt[-(1/(M^2*\[Kappa]^2*(1 + Log[\[
```

5.6 Solved constraints

```
"Skipped generic symbolic solve in the remote d<=6 run to avoid kernel OOM; the full sector-by-
```

6 Differences

6.1 Supported difference

```
<|{1, 0} -> {-1/2*(KappaRefFinal*LambdaRefFinal) + (Sqrt[3/2]*Sqrt[-(1/(M^2*\[Kappa]^2*(1 + Log
```

6.2 Unsupported difference

```
0
```

6.3 Independent quadratic cross-check

A second script verified the completed {2,4} statement directly from the saved vectors, without using the main analyzer helper.

```
<|"PureRedefinitionHasSolutionQ" -> True, "PureRedefinitionReconstructionMatchesQ" -> True, "Jo
```

7 Conclusion

An exact match exists after imposing the verified matching substitutions and applying the enlarged quadratic quotient, including the completed treatment of the {2,4} sector. The independent saved-vector cross-check confirms that the remaining quadratic four-derivative sector is a pure quadratic field-redefinition image with vanishing curvature-squared coefficients.